

Modeling Cell-Cell Interactions in T-Cell Dependent Cytotoxicity (TDCC) Based Cancer Therapeutics

Takeda Oncology

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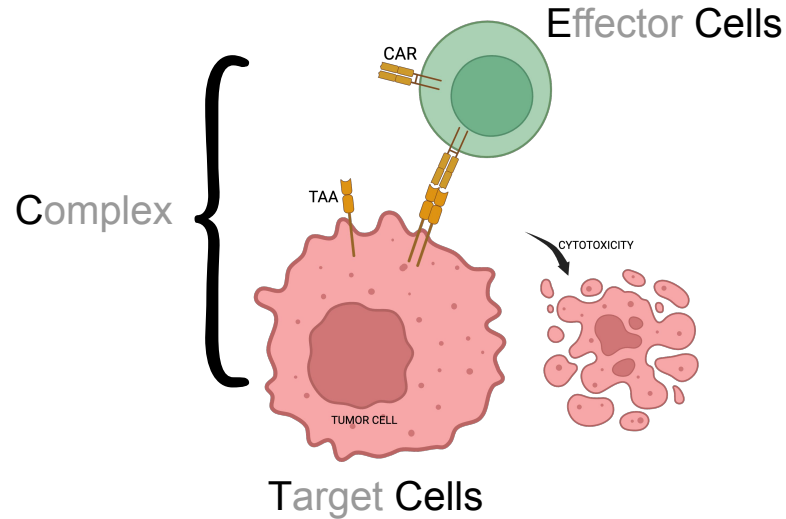
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Background

CAR T Cells (Chimeric Antigen Receptors) TAA (Tumor-Associated Antigen)



Goals

- Is the **E:T ratio** a reliable independent variable?
- Does cytotoxicity remain **invariant** across identical E:T ratios?
- If not, what are the **unaccounted variables**?



Overview:



Start with ABM

Reflect cell-level behavior

Align with observable
lab dynamics

Bridge to ODE

Fit to ABM dynamics

Study the goal efficiently

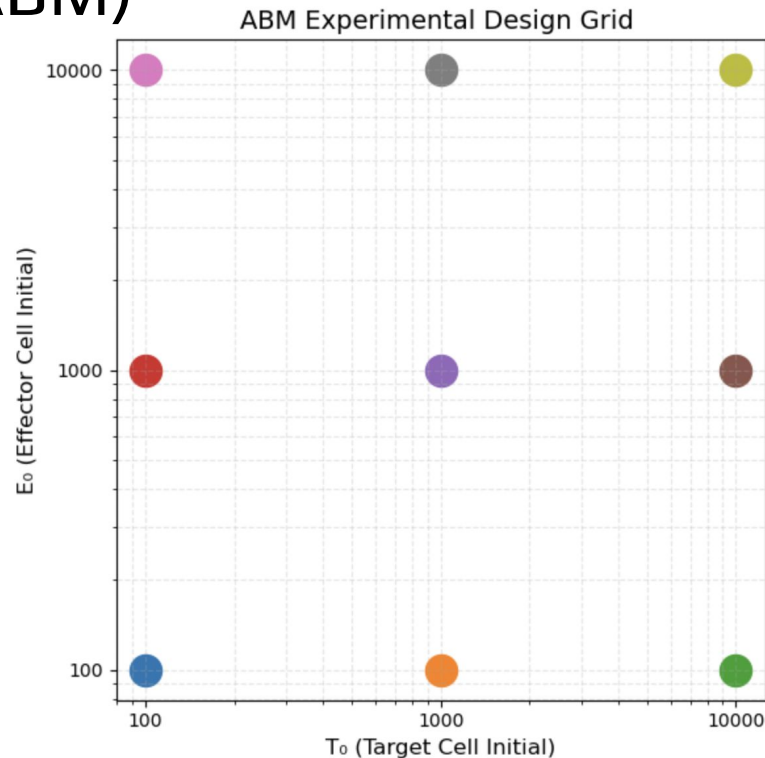
Add Master Equation

Track probability distribution

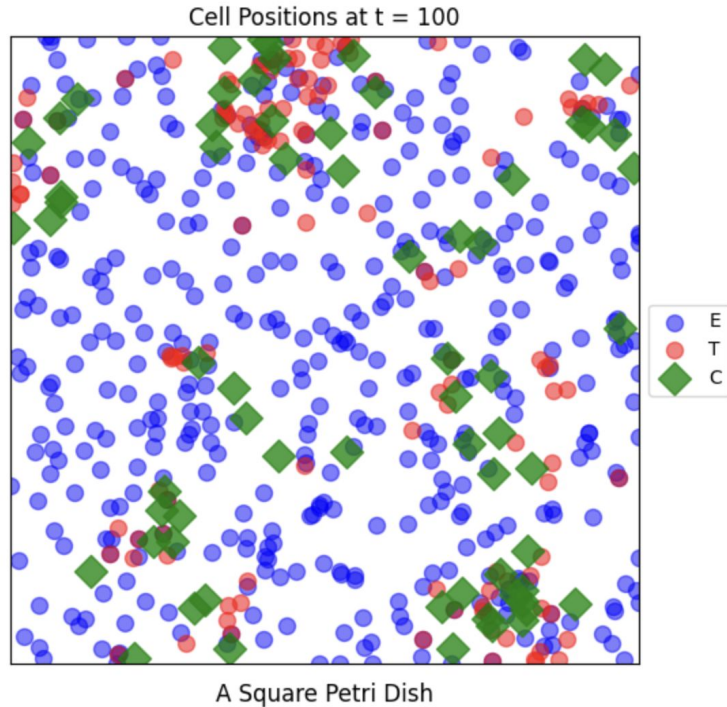
Understand rare outcomes

Experimental Design Grid (ABM)

- Full factorial grid of initial conditions
- 9 unique (E_0, T_0) combinations
- Basis for simulation of cell interaction dynamics



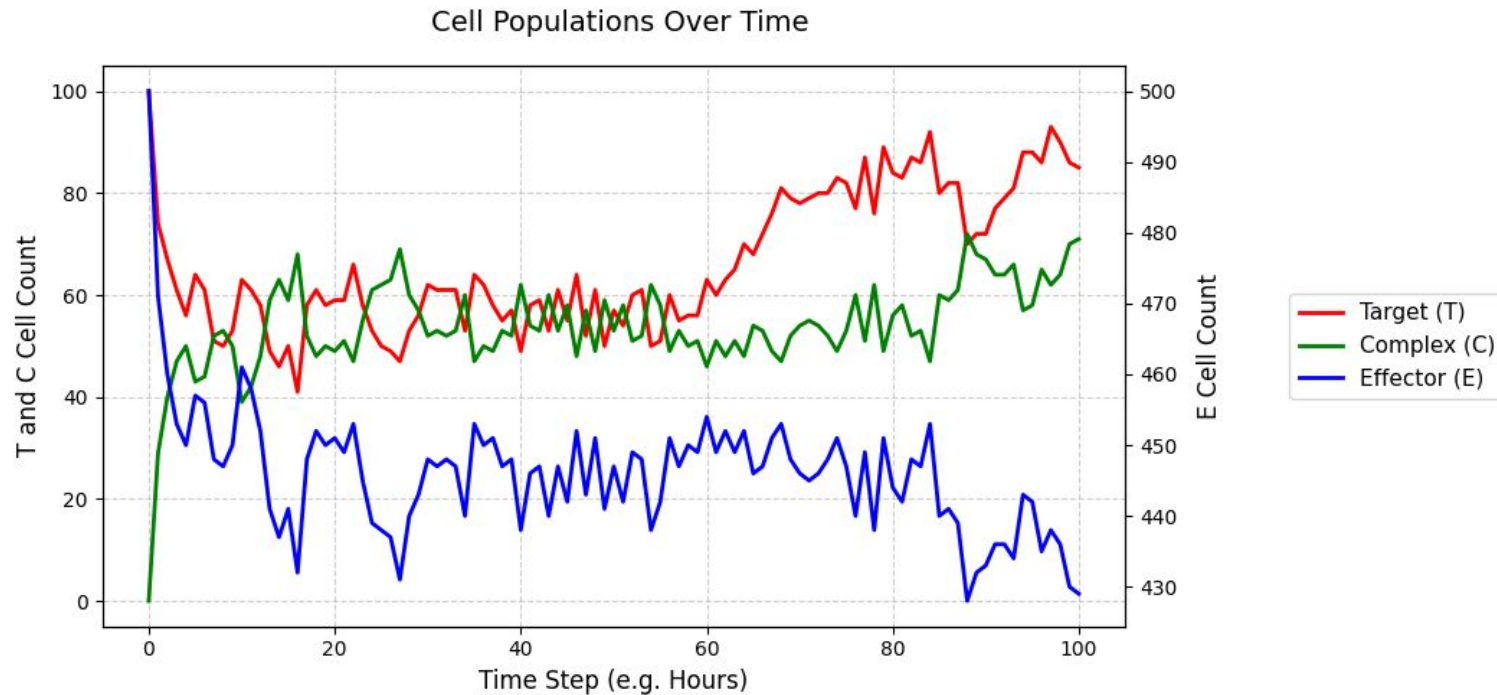
Agent-Based Model



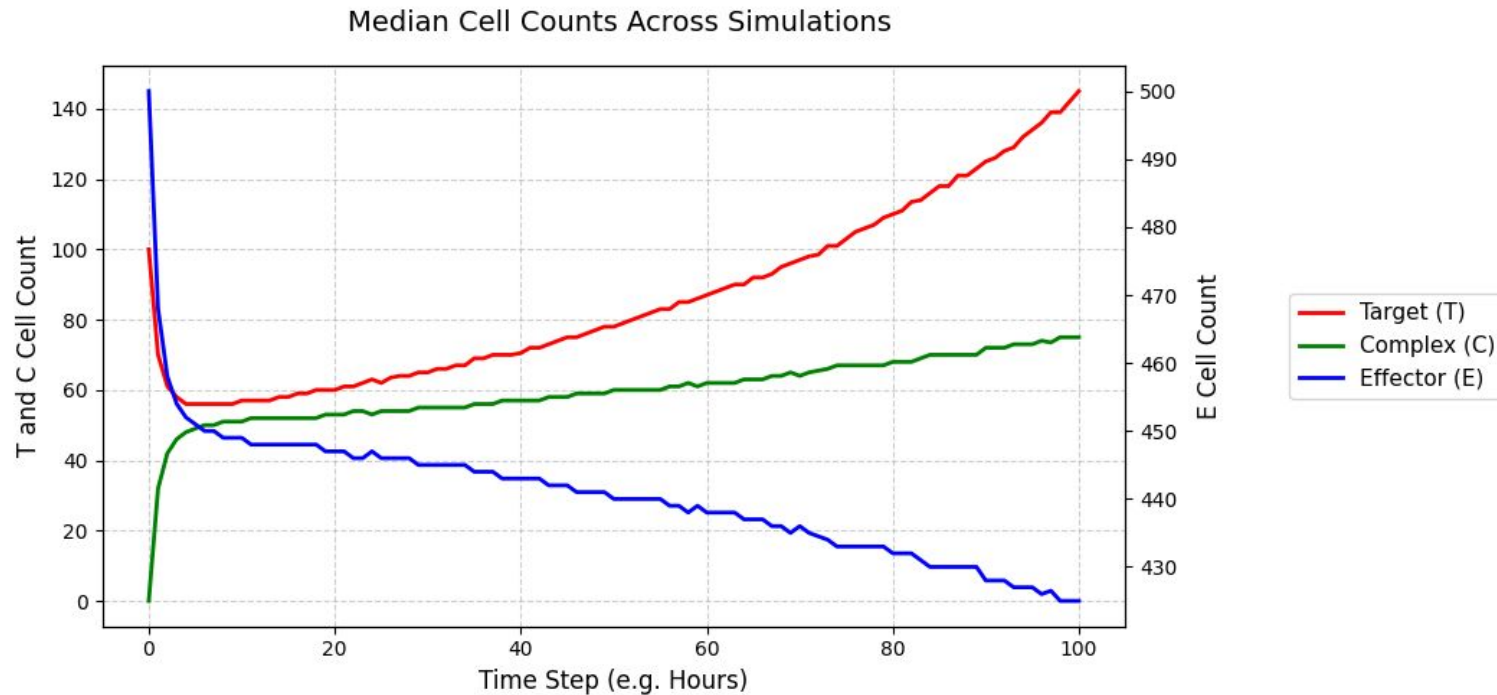
Replicate Cell-Level Behaviors

- Effectors move freely; bind and kill targets upon contact
- Full control over binding, killing, and movement rates

Agent-Based Model



Agent-Based Model



Record the trajectories...

Time Step	Effector (E)	Target (T)	Complex (C)
0	500	100	0
1	468	70	32
2	458	61	42
3	454	58	46
4	452	56	48
...	...	...	...
96	426	136	74
97	427	139	74
98	425	139	75
99	425	142	75
100	425	145	75

Now, we can use ODE to fit this data!



Other Progress and Exploration



- ABM exploration
 - Captures spatial & stochastic dynamics
 - Real cell interactions more complex
 - Current ABM simplifies real tumor-immune behavior



ODE Recall

In the Life cycle diagram:

E : Effector cell

T : Target cell

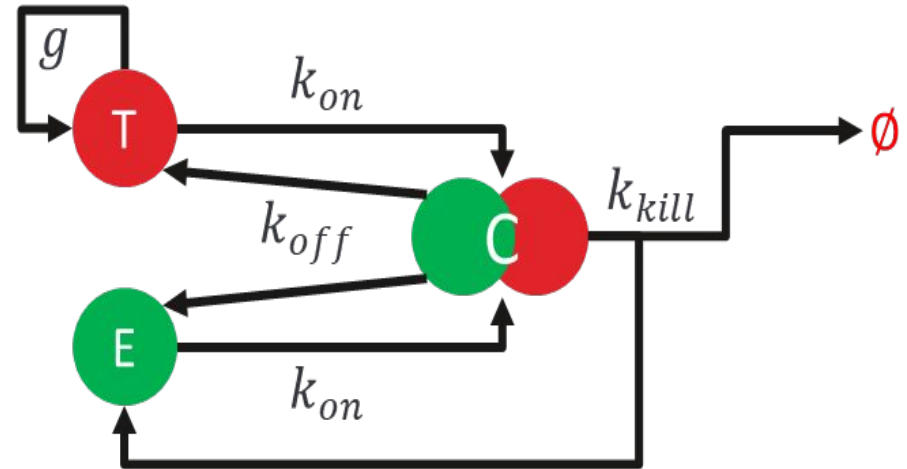
C: E+T: Complex cell

k_{on} - Binding rate constant

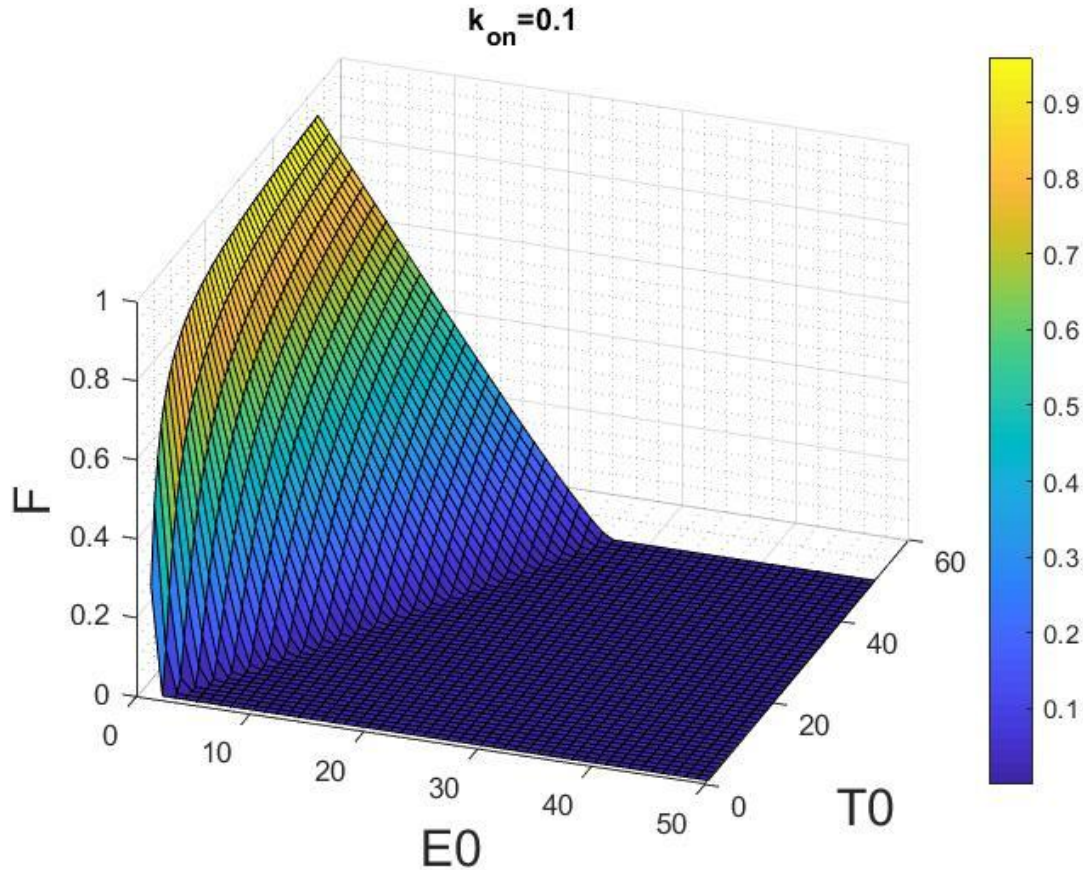
k_{off} - Unbinding rate

k_{kill} - Killing rate

g - Growth rate



ODE F Surface plot

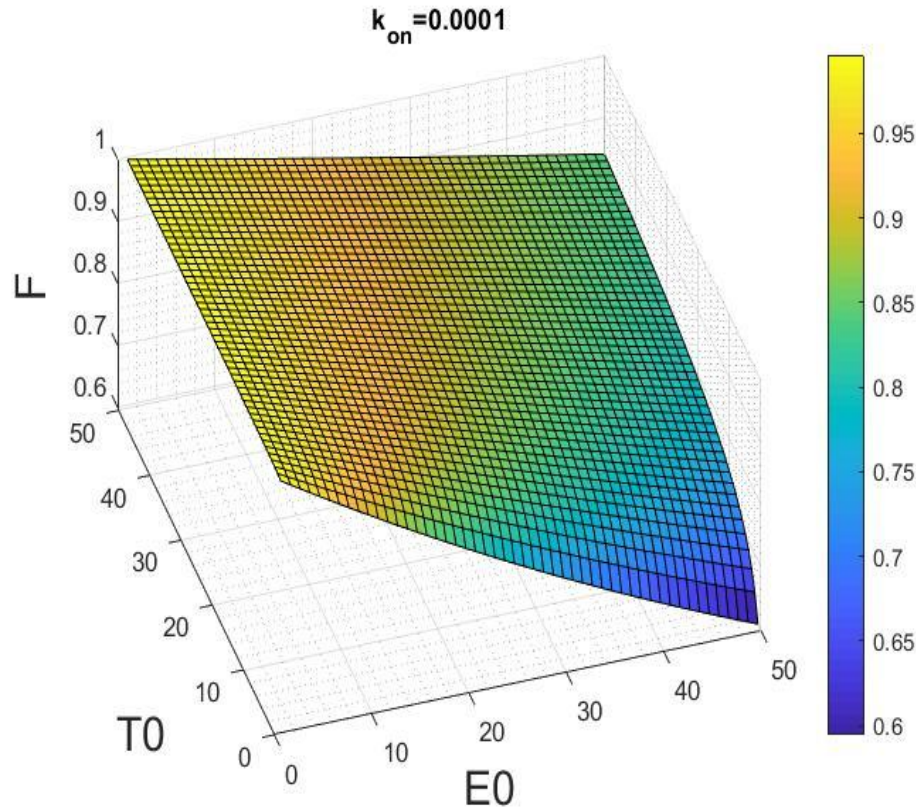


$$F(t) = \frac{T(t, E_0)}{T(t, 0)} = \frac{\text{Tumor with Treatment}}{\text{Tumor without Treatment}}$$

E:T ratio

- Parameters are:
 $g = 0.09$; $k_{on} = 0.1$; $k_{off} = 0.3$; $k_{kill} = 0.1$
- F is zero in the part below the diagonal for higher Effector than Target cell.

ODE F Surface plot



Other variable:

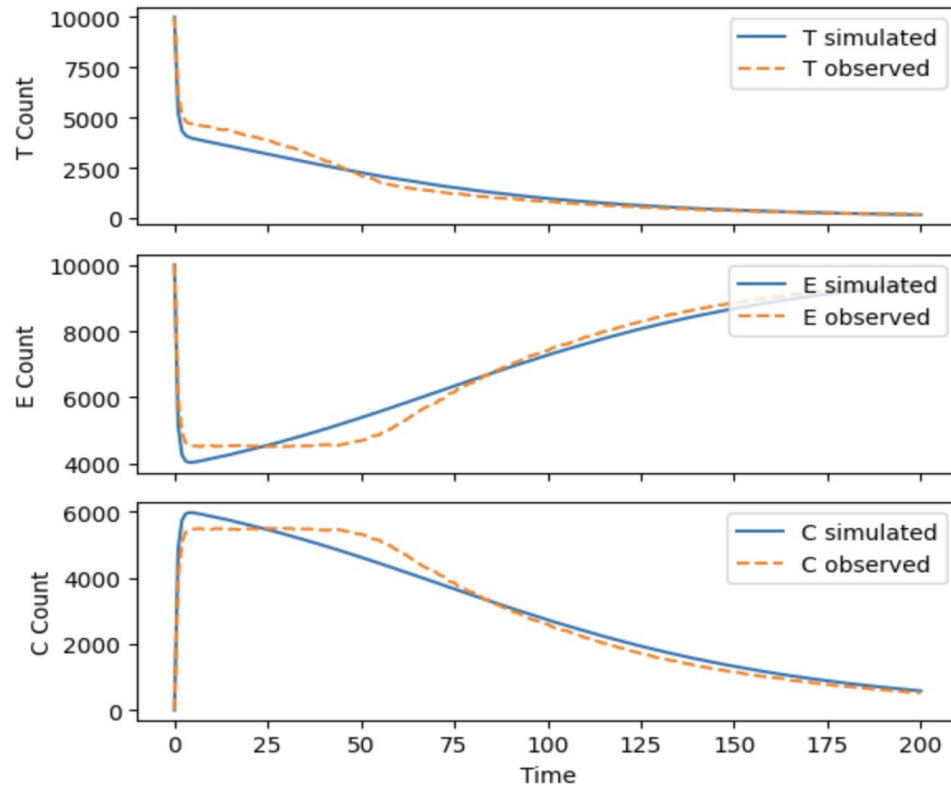
- Binding rate is significant in Tumor cytotoxicity.

ODE Fitting

≡ fitting $E_0 = 10000$, $T_0 = 10000$ now ≡

Best k_{on} : 0.0001, original k_{on} was 0.4, 3272.0 times

ODE Fit with $k_{on} = 0.0001$

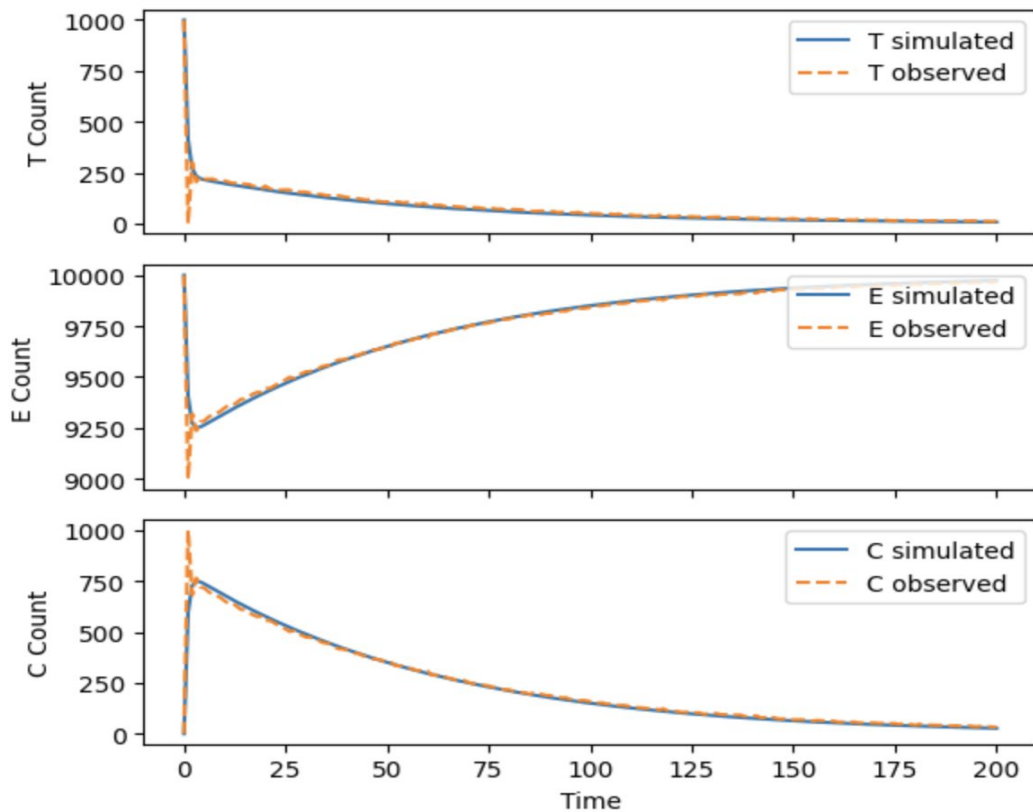


ODE Fitting

≡ fitting $E_0 = 10000$, $T_0 = 1000$ now ≡

Best k_{on} : 0.0001, original k_{on} was 0.4, 3429.0 times

ODE Fit with $k_{on} = 0.0001$



Master Equation

“How the chance of having a certain number of cells changes over time.”

$$P'_n(t) = -(\lambda + \mu)nP_n(t) + \lambda(n-1)P_{n-1}(t) + \mu(n+1)P_{n+1}(t)$$



Outflow from current state due
to either proliferation or death



Inflow from smaller state
by one cell replicating



Inflow from larger state
by one cell dying

$P_n(t)$: Probability of having n T cells at time t .



Master Equation

“How the chance of having a certain number of cells changes over time.”

$$P'_n(t) = -(\lambda + \mu)nP_n(t) + \lambda(n - 1)P_{n-1}(t) + \mu(n + 1)P_{n+1}(t)$$

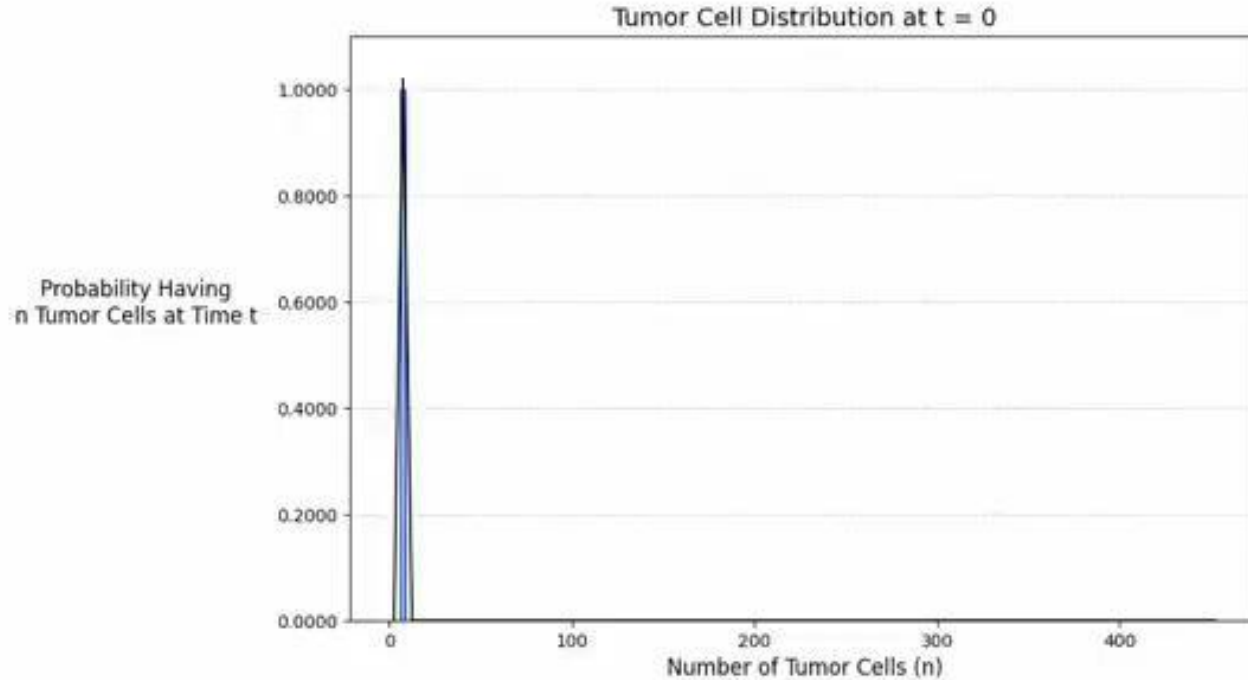
Mean $N_0 \exp((\lambda - \mu)t)$

Variance $N_0 \exp(2(\lambda - \mu)t) \{1 - \exp((\mu - \lambda)t)\} \frac{\lambda + \mu}{\lambda - \mu}$

$P_n(t)$: Probability of having n T cells at time t .



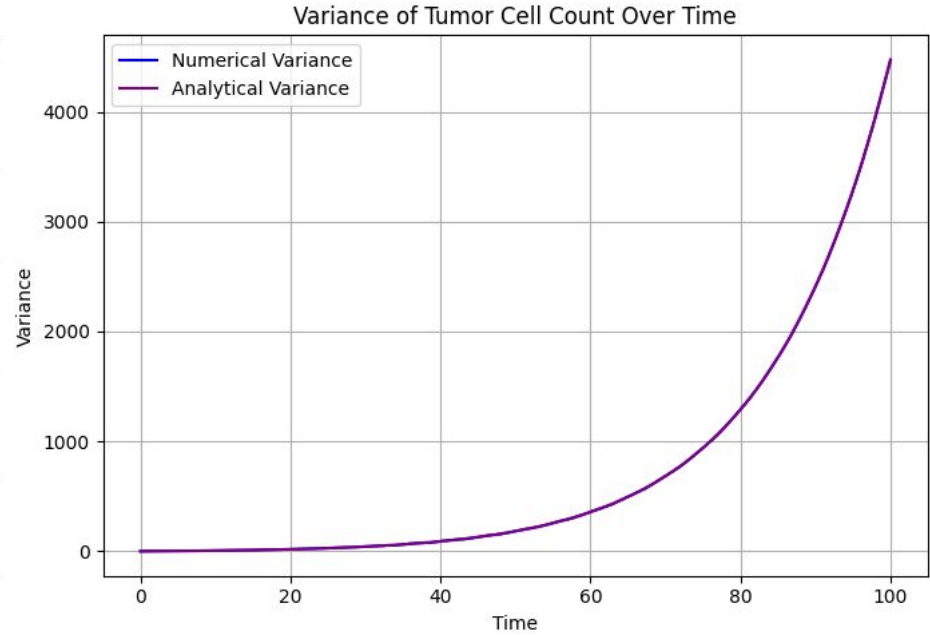
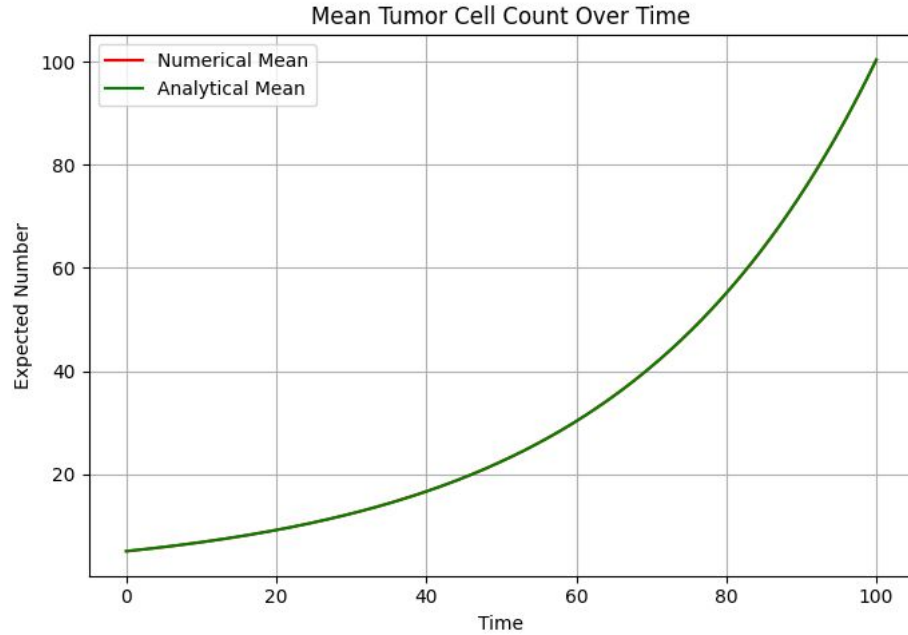
Master Equation



Why probability of having 0 tumor cell is a little high towards the end?



Expected number of Tumor cells and its variance



Numerical and analytical results match.



Other Progress and Exploration



- 2 Dimensional Master Equation (model both E and T cells)
 - Can be setup
 - Can be solved numerically
 - Can be generalized to 3 Dimensional (also model C cells)
 - But...
 - Very computationally costly!
- Everything is nice, linear and smooth...
 - Which is a bad thing!
 - Rare event does not arise
 - Not really adding values to existing ABM and ODE models

Next Steps

- Master equation analysis
- Fit ABM to real data
- Compare E:T ratio across models
- Explore ODE-based results



Thank you !

